

# Hein Zarni Aung

+1-(917)-854-7709 | heinzarniaung.dev@gmail.com | linkedin.com/in/heaung | github.com/LoistCloist

## EDUCATION

### Stony Brook University

Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Software Security, Systems Fundamentals I & II, Algorithms, Statistics & Probability

## INTERNSHIPS

### Software Engineering Intern — Clustr

Jan 2026 - Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.
- Designed and implemented a Scheduler UI module featuring tabbed navigation, a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

### Software Engineering Intern — Prenora Dynamics LLC

Jun 2026 - Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, and built a seed-data workflow to streamline debugging and local development.

## RESEARCH

### Undergraduate Research Assistant — Ethos Lab, Stony Brook University

Sep 2025 - Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag fraudulent domains impersonating EZ-Pass tolling services, using Selenium to automate certificate/domain scraping and Python to filter and triage flagged sites via keyword- and heuristic-based detection logic developed with a two-person team.

## CAMPUS INVOLVEMENT

### Teaching Assistant — Data Structures & Algorithms, Stony Brook University

Aug 2025 - Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

## PROJECTS

### TuneAcademy

React, NumPy, Essentia, Firestore

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances; pivoted from a planned ML model to rule-based scoring under hackathon time/compute constraints.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy to Cloudflare Workers, with the broader app spanning a React/Tailwind frontend, Python/FastAPI backend in Docker, Cloud Firestore, and GCP hosting.

## SKILLS

**Languages:** Java, Python, JavaScript, TypeScript, C

**Frontend:** React, React Native, Next.js

**Backend:** Node.js, Express.js, FastAPI

**Databases & Cloud:** PostgreSQL, Firebase/Firestore, MongoDB, AWS, GCP

**Tools:** Git, Docker, Linux, PM2, Selenium